

MAWLANA BHASHANI SCIENCE AND TECHNOLOGY UNIVERSITY

SANTOSH, TANGAIL-1902



DEPARTMENT OF INFORMATION AND COMMUNICATION TECHNOLOGY

Lab Report

Lab Report No : 05

Lab Report Name : Star Topology Establishment Using a Switch & Straight Cable

Course Title : Computer Networks Lab

Course Code : ICT-3202

Submitted By	Submitted To
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Date of Performance :

Date of Submission :

Experiment No: 05

Experiment Name: Star Topology Establishment Using a Switch & Straight Cable

Objectives:

1. To understand the principles and advantages of star topology networking
2. To establish a local area network using a network switch as the central connection point
3. To configure IP addresses for multiple computers on the same network
4. To verify connectivity between all computers in the star topology
5. To enable file sharing among multiple computers in a star network topology
6. To demonstrate internet access through a gateway in a star topology configuration
7. To implement proper security and network profile settings for multiple computers in a networked environment

Necessary Materials and Software:

1. Cat6 Straight-Through Cables (Three)
2. Network Switch + power supply for the switch
3. Three PCs/Laptops, each with a functional Ethernet (RJ45) network interface card
4. Windows 10/11 Operating System
5. Windows Settings / Control Panel (for IP configuration and network profile setup)
6. Windows Command Prompt (Administrator access)
7. File Explorer

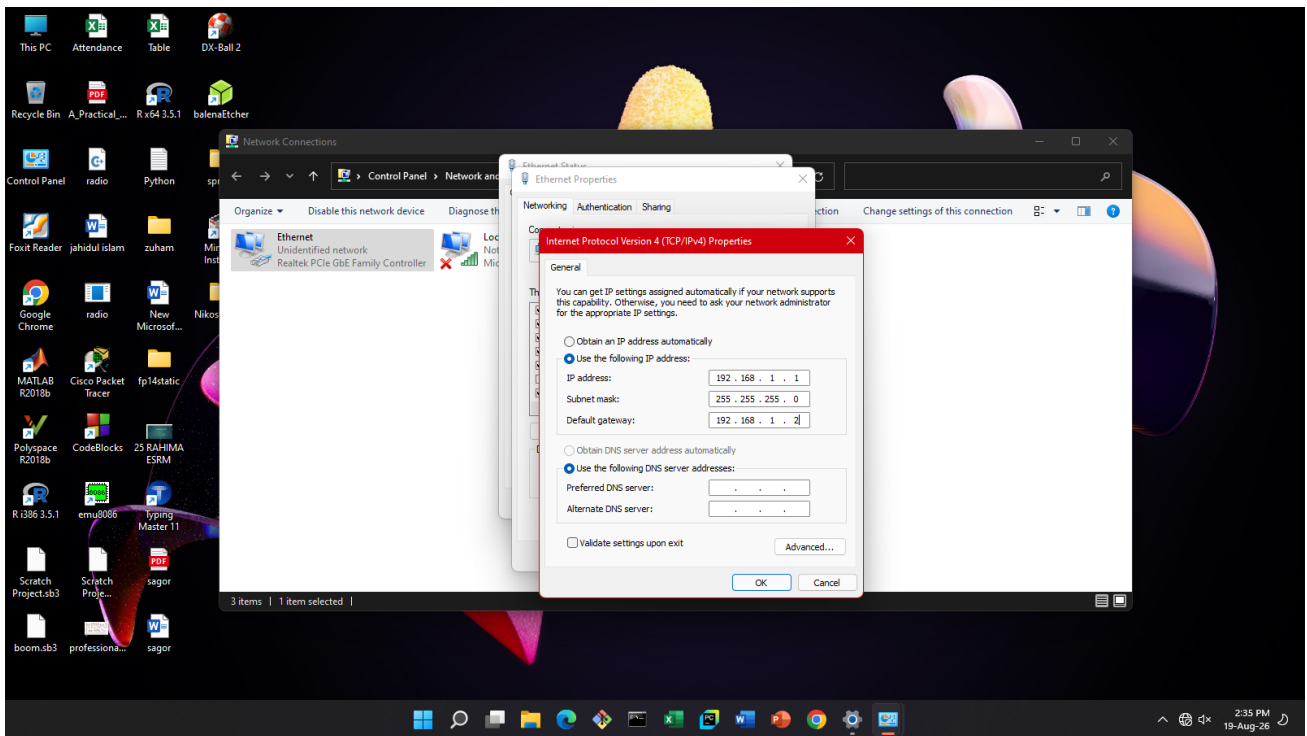
Procedure:

Step 1: Physical Connection

1. Place the network switch in a central location
2. Connect Computer 1 to any port on the switch using a straight cable
3. Connect Computer 2 to another port on the switch using a straight cable
4. Connect Computer 3 to another port on the switch using a straight cable
5. Power on the switch and verify that the LED indicators are active on all connected ports
6. Ensure all computers are powered on

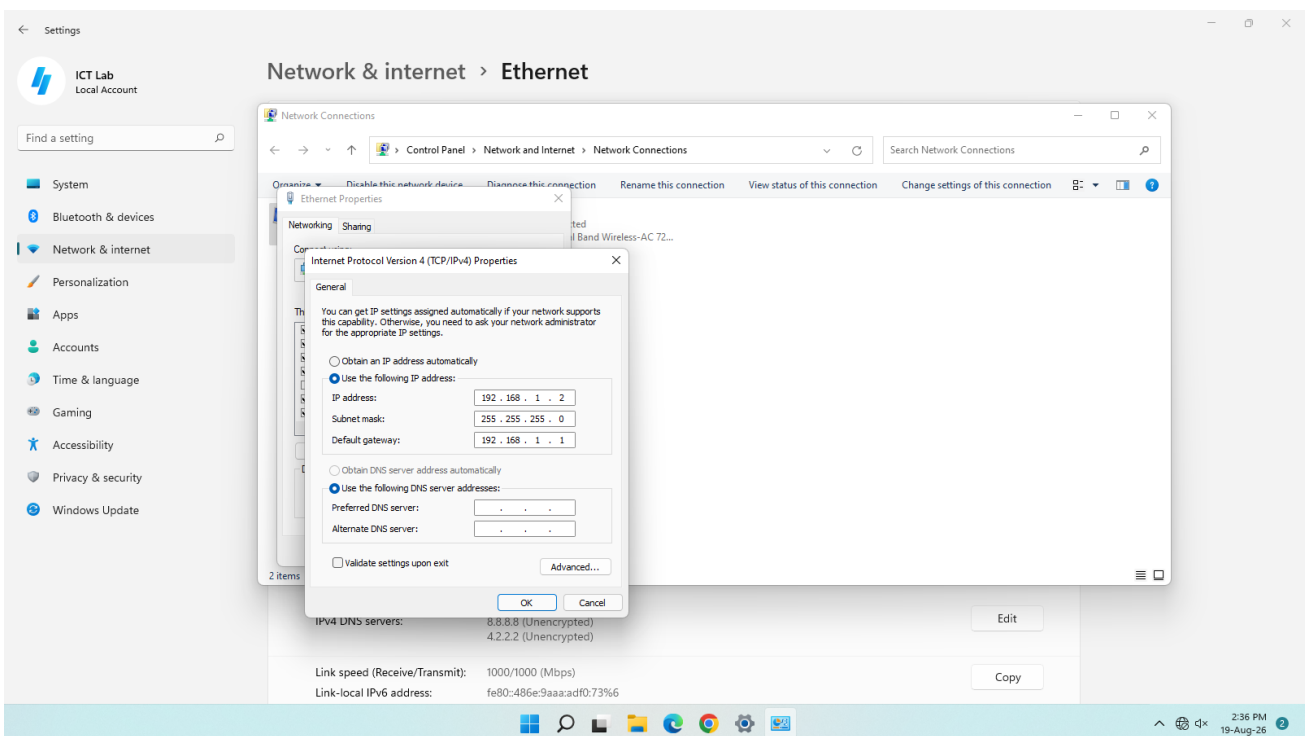
Step 2: IP Configuration for Computer 1 (Local Network)

1. Open Settings from the Start menu
2. Navigate to Network & Internet
3. Click on Ethernet, then Select Properties
4. Click on Internet Protocol Version 4 (TCP/IPv4)
5. Click Properties
6. Configure the following settings:
7. IP Address: 192.168.1.1
8. Subnet Mask: 255.255.255.0
9. Click OK to apply changes



Step 3: IP Configuration for Computer 2 (Local Network)

1. Repeat the same process as Step 2 on Computer 2
2. Configure the following settings:
3. IP Address: 192.168.1.2
4. Subnet Mask: 255.255.255.0
5. Click OK to apply changes

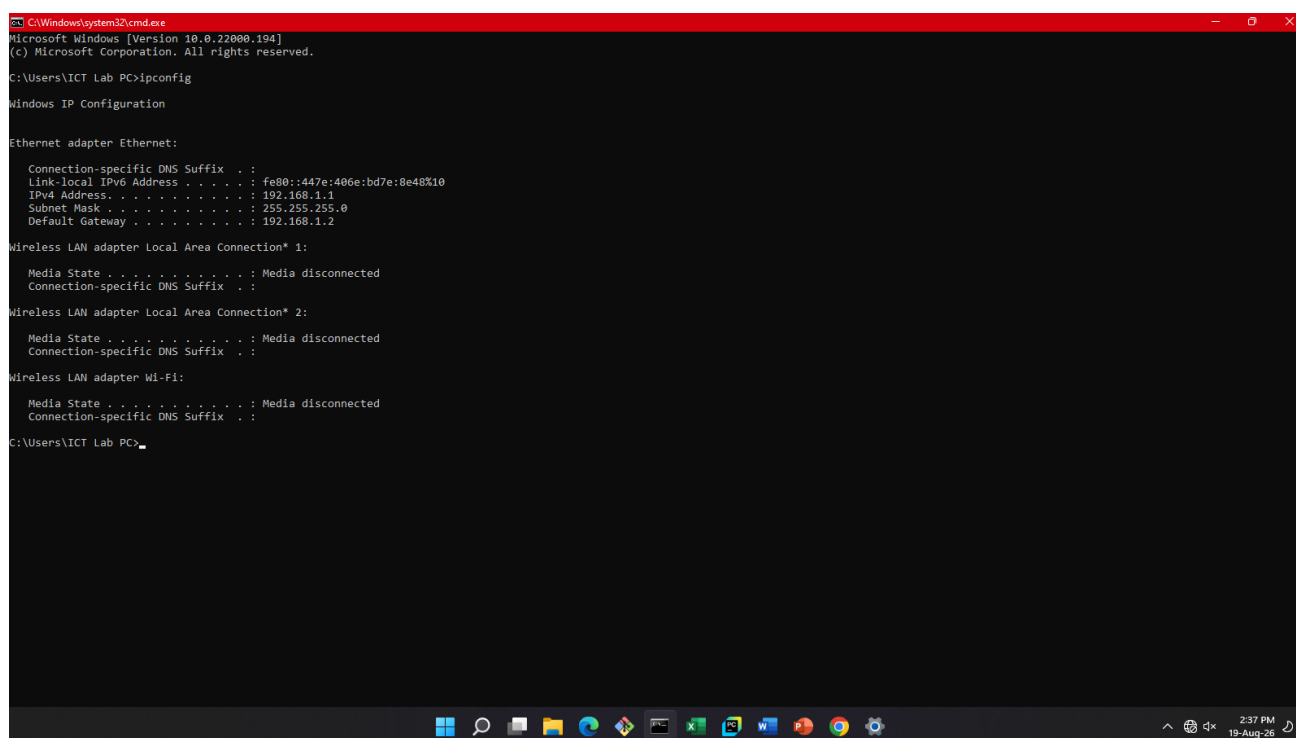


Step 4: IP Configuration for Computer 3 (Local Network)

1. Repeat the same process as Step 2 on Computer 3
2. Configure the following settings:
3. IP Address: 192.168.1.3
4. Subnet Mask: 255.255.255.0
5. Click OK to apply changes

Step 5: Verify Network Connectivity

1. Open Command Prompt on Computer 1
2. Ping Computer 2: ping 192.168.1.2
3. Ping Computer 3: ping 192.168.1.3
4. Verify that replies are received from both computers
5. Repeat ping tests from Computer 2 and Computer 3 to verify bidirectional communication with all other computers
6. Document successful connectivity results



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C:\Windows\system32\cmd.exe
Microsoft Windows [Version 10.0.22000.194]
(c) Microsoft Corporation. All rights reserved.

C:\Users\ICT Lab PC>ipconfig

Windows IP Configuration

Ethernet adapter Ethernet:

    Connection-specific DNS Suffix  . : 
    Link-local IPv6 Address . . . . . : fe80::447e:406e:bd7e:8e48%10
    IPv4 Address. . . . . : 192.168.1.1
    Subnet Mask . . . . . : 255.255.255.0
    Default Gateway . . . . . : 192.168.1.2

Wireless LAN adapter Local Area Connection* 1:

    Media State . . . . . : Media disconnected
    Connection-specific DNS Suffix  . : 

Wireless LAN adapter Local Area Connection* 2:

    Media State . . . . . : Media disconnected
    Connection-specific DNS Suffix  . : 

Wireless LAN adapter Wi-Fi:

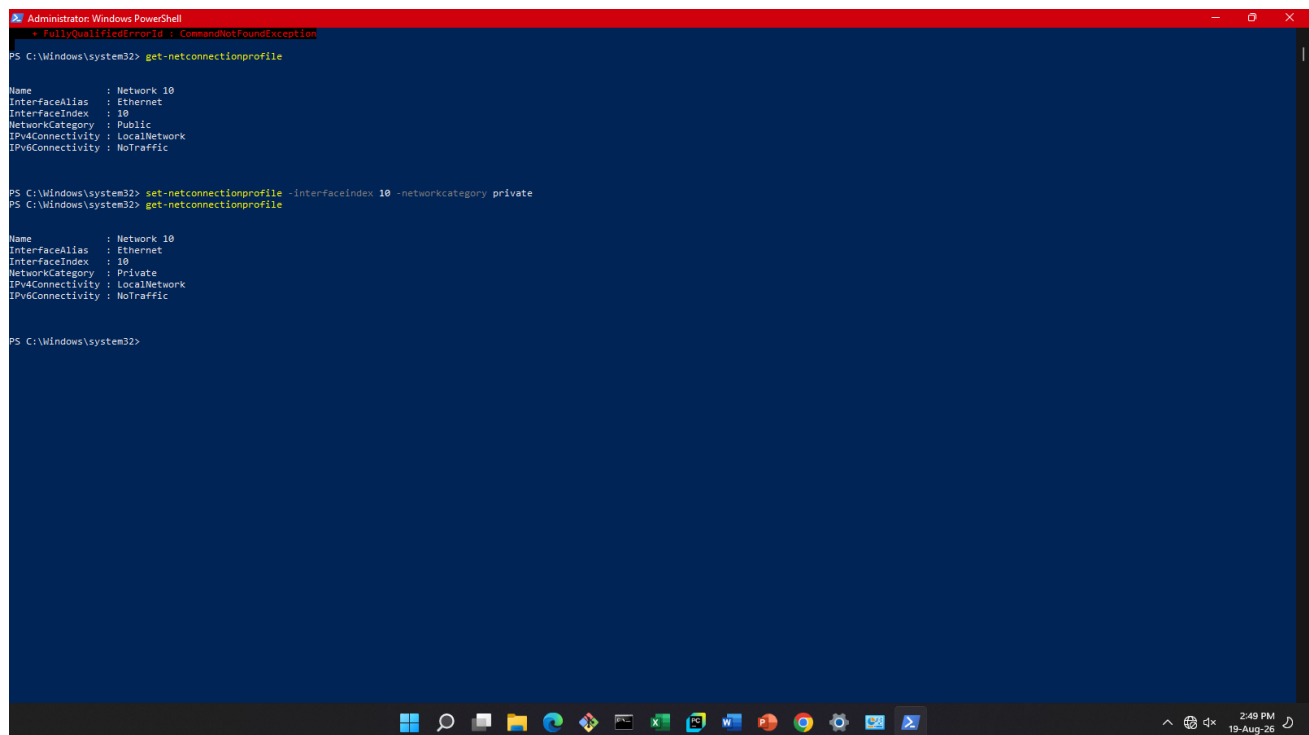
    Media State . . . . . : Media disconnected
    Connection-specific DNS Suffix  . : 

C:\Users\ICT Lab PC>
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Step 6: Set Network Profile to Private on All Computers On Computer 1:

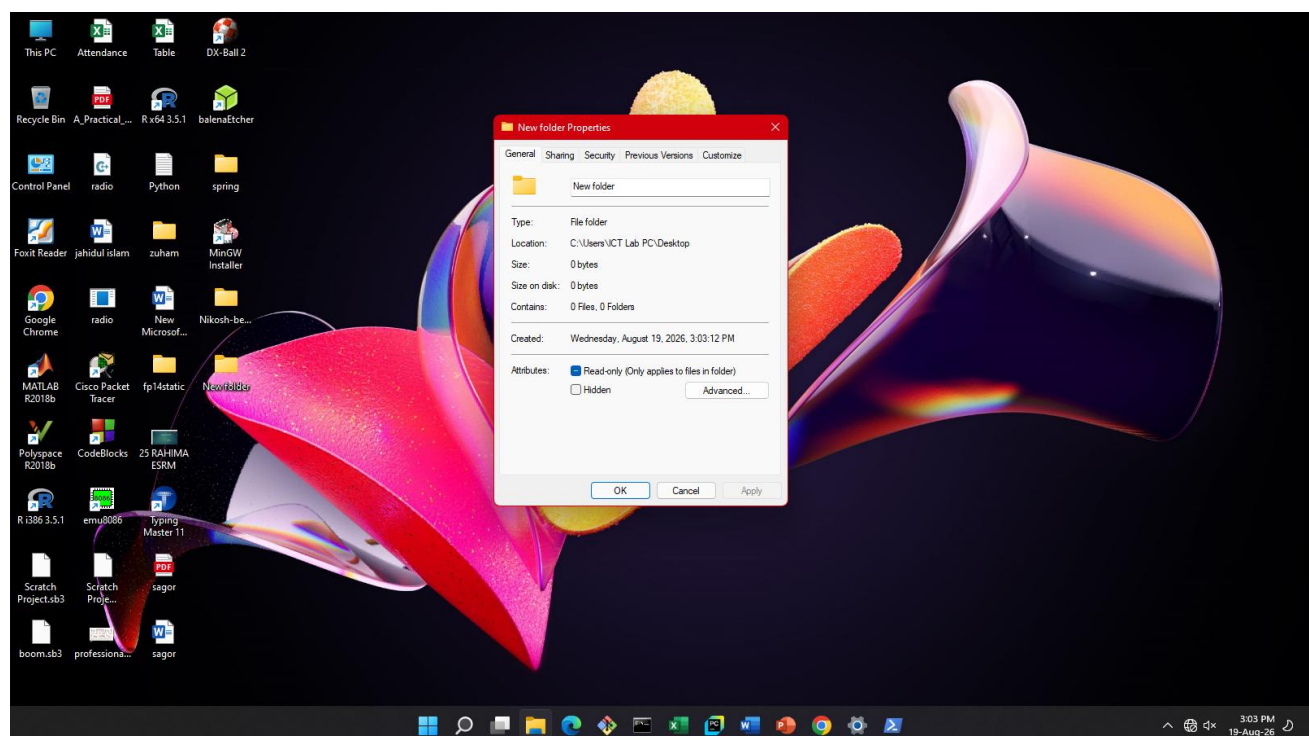
1. Open Settings (Win + I)
2. Click Network & Internet
3. Click on Ethernet (your active connection)
4. Under Network profile type, select Private
5. Alternatively, using PowerShell as Administrator:
6. Run: Get-NetConnectionProfile
7. Run: Set-NetConnectionProfile -Name "Ethernet" -NetworkCategory Private

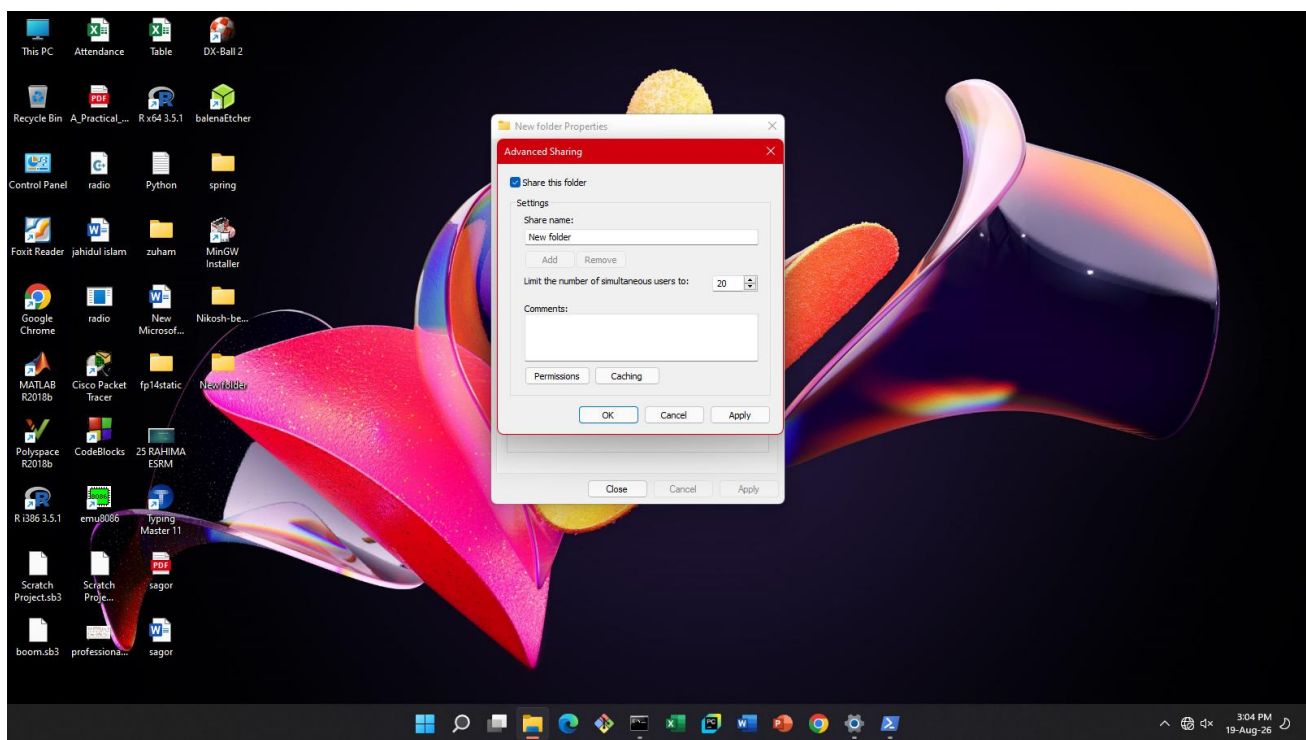
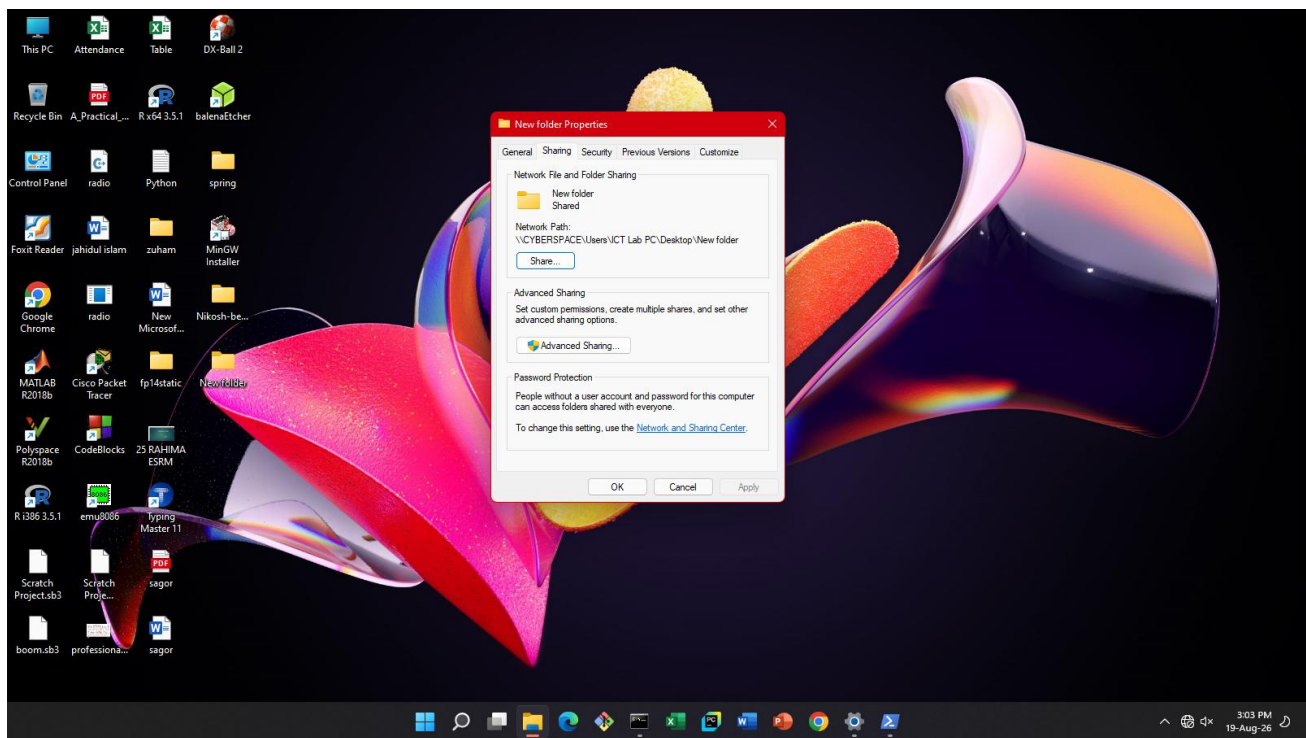
Repeat Step 6 on Computer 2 and Computer 3 using the same settings.

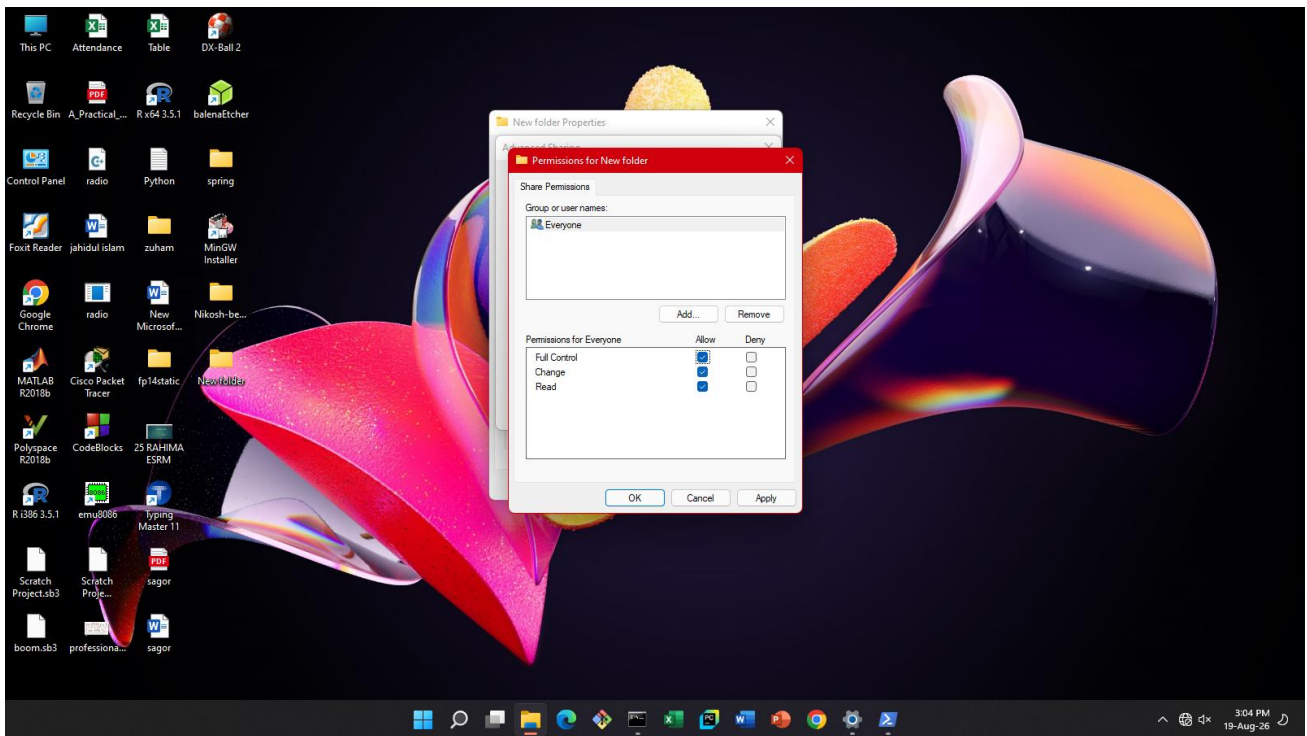


Step 7: Enable File Sharing on Computer 1

1. Create or select a folder you want to share
2. Right-click on the folder and select Properties
3. Go to the Sharing tab
4. Click Advanced Sharing
5. Check the box "Share this folder"
6. Click Apply, then OK

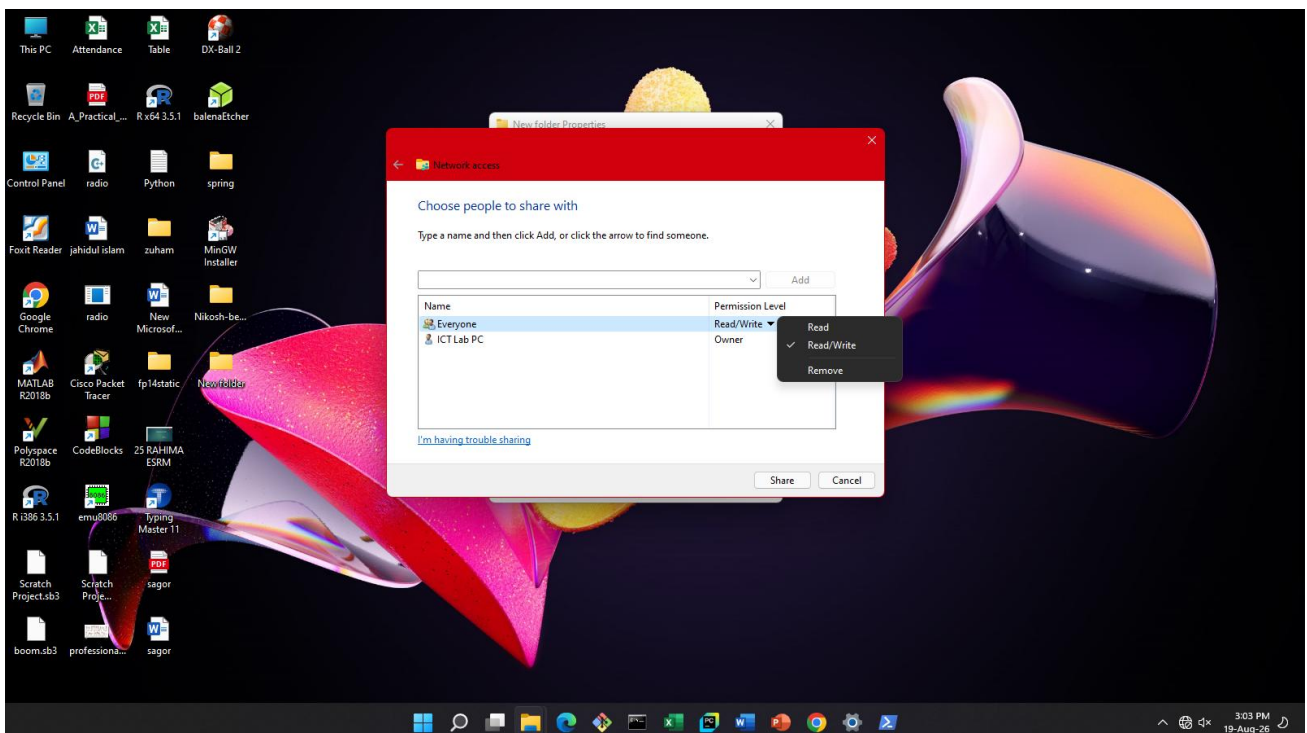






Step 8: Configure Security Access on Computer 1

1. In the folder Properties, go to the Security tab
2. Click Edit under "Group or user names"
3. Click Add
4. Type "Everyone" and click Check Names ND Click OK
5. Select Everyone and grant full permissions (Read, Write, etc.)
6. Click Apply, then OK

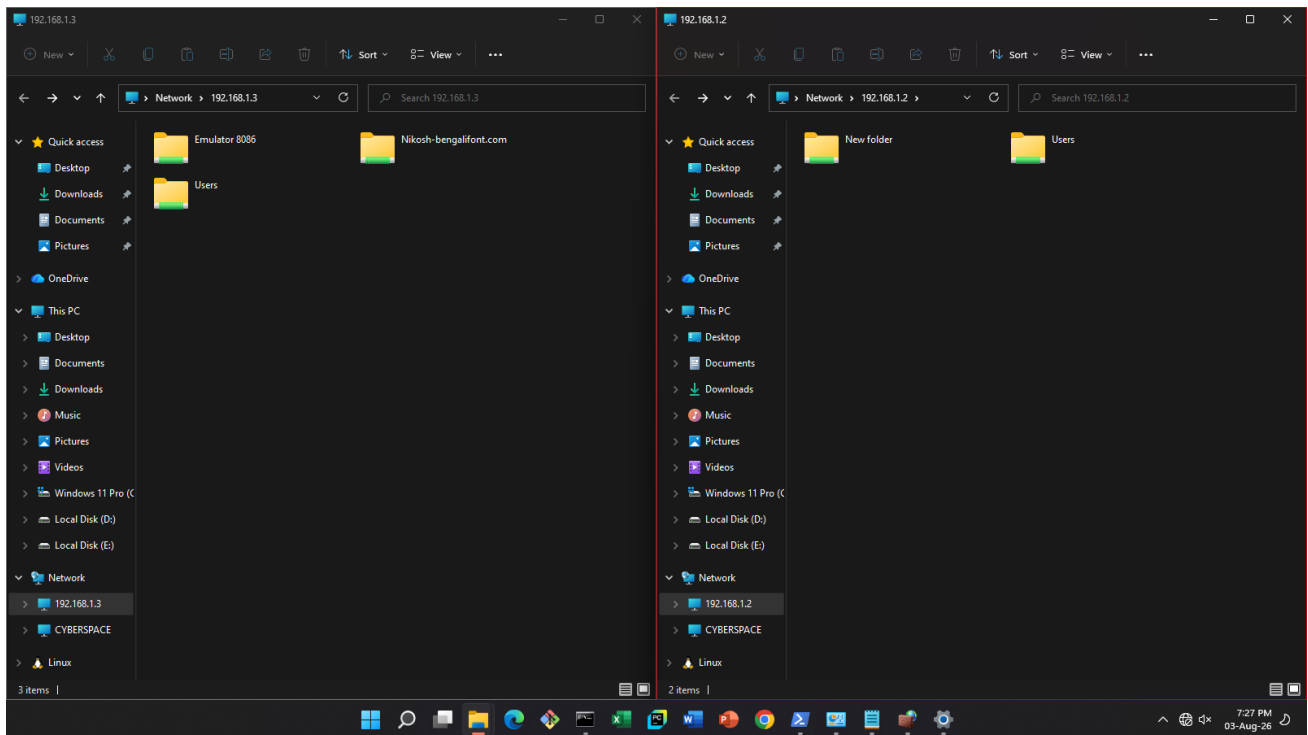


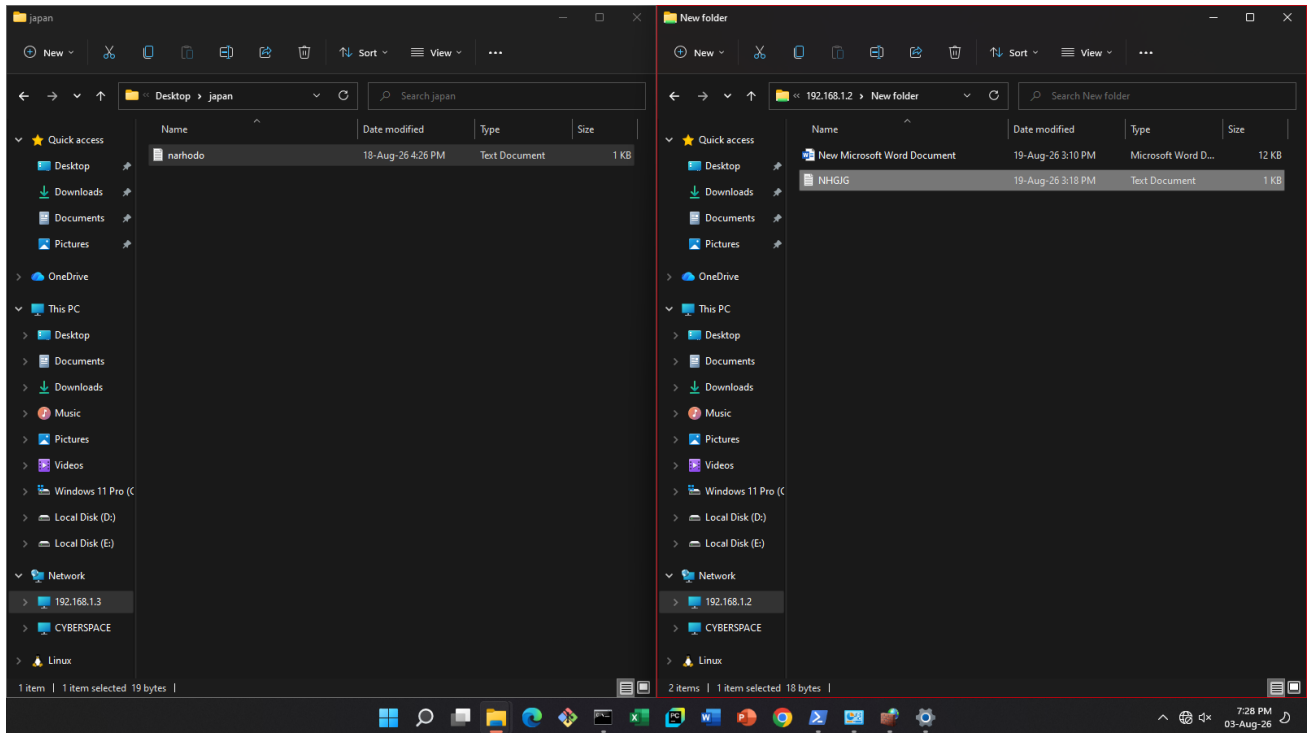
Step 9: Enable File Sharing on Computer 2 and Computer 3

1. Repeat Steps 7 and 8 on Computer 2 and Computer 3
2. Create shared folders on both computers for testing

Step 10: Test File Sharing from Computer 1

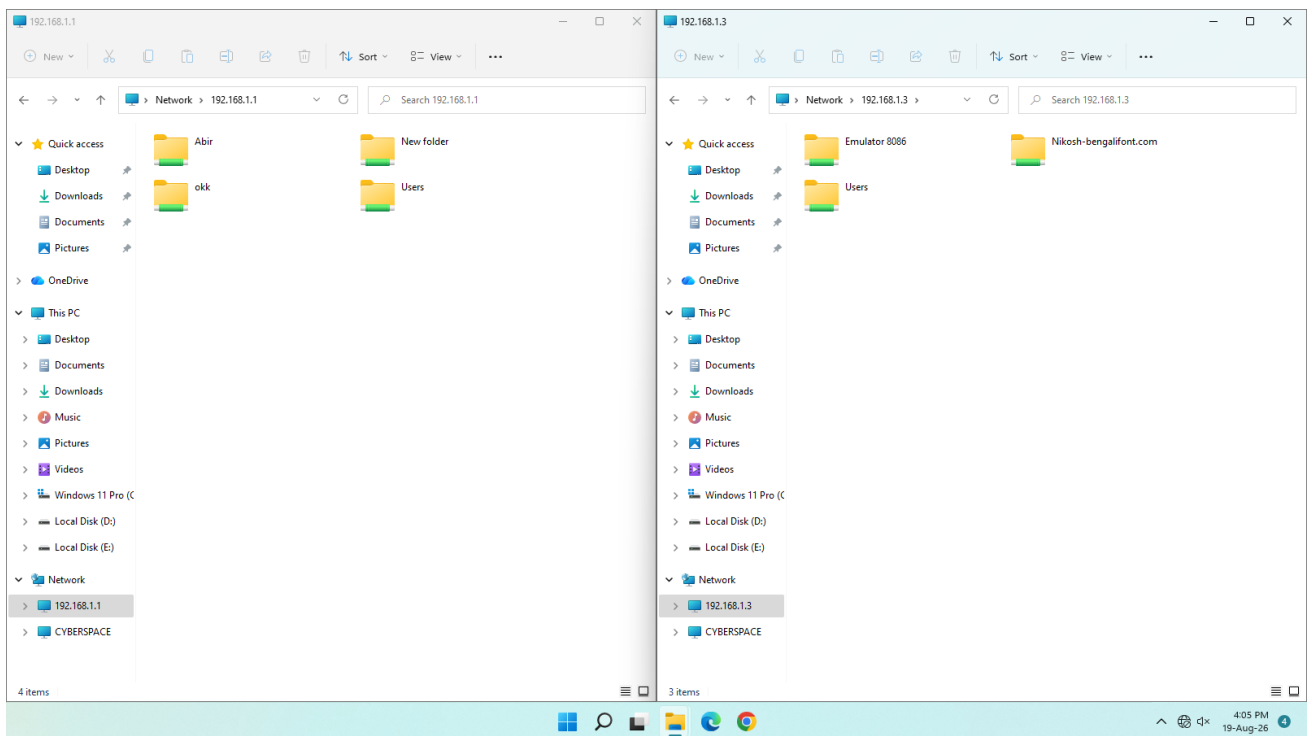
1. Open File Explorer on Computer 1
2. In the address bar, type: \192.168.1.2
3. Open the shared folder from Computer 1
4. Copy a test file to verify write access
5. Repeat this for Computer 3's shared folder: \192.168.1.3

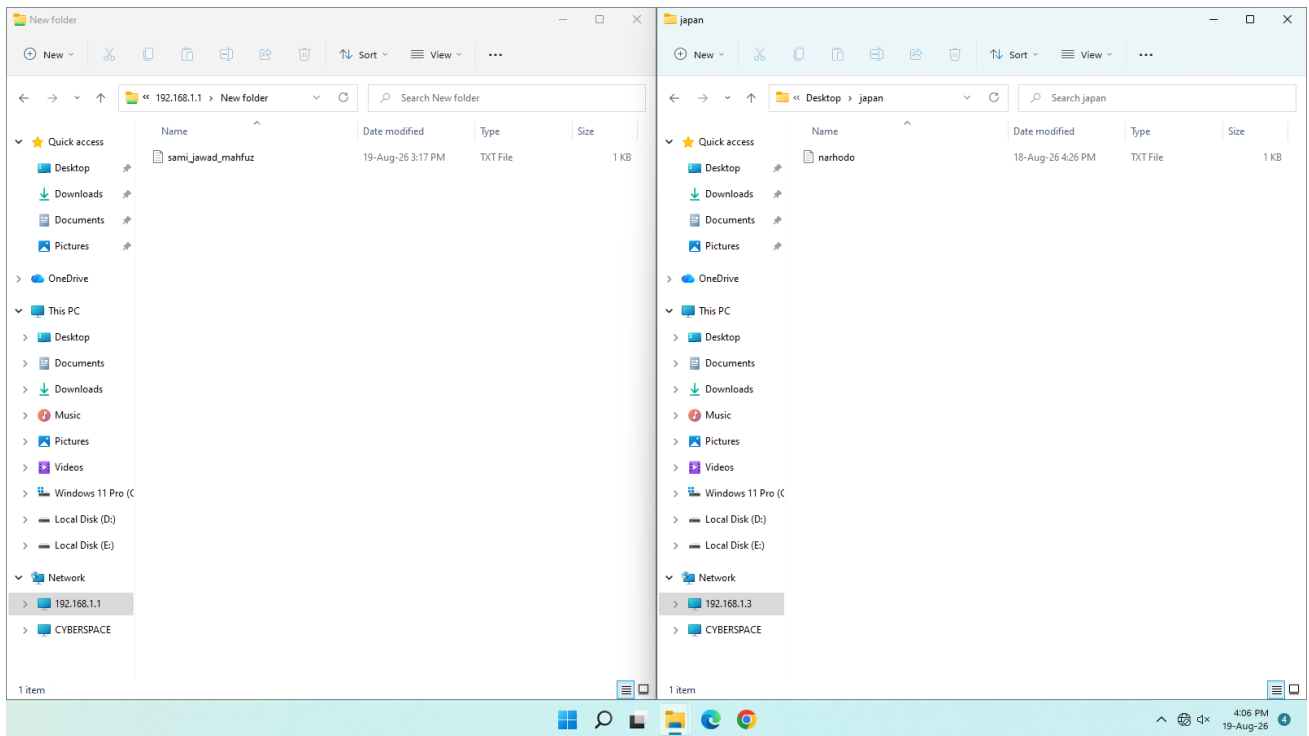




Step 11: Test File Sharing from Computer 2

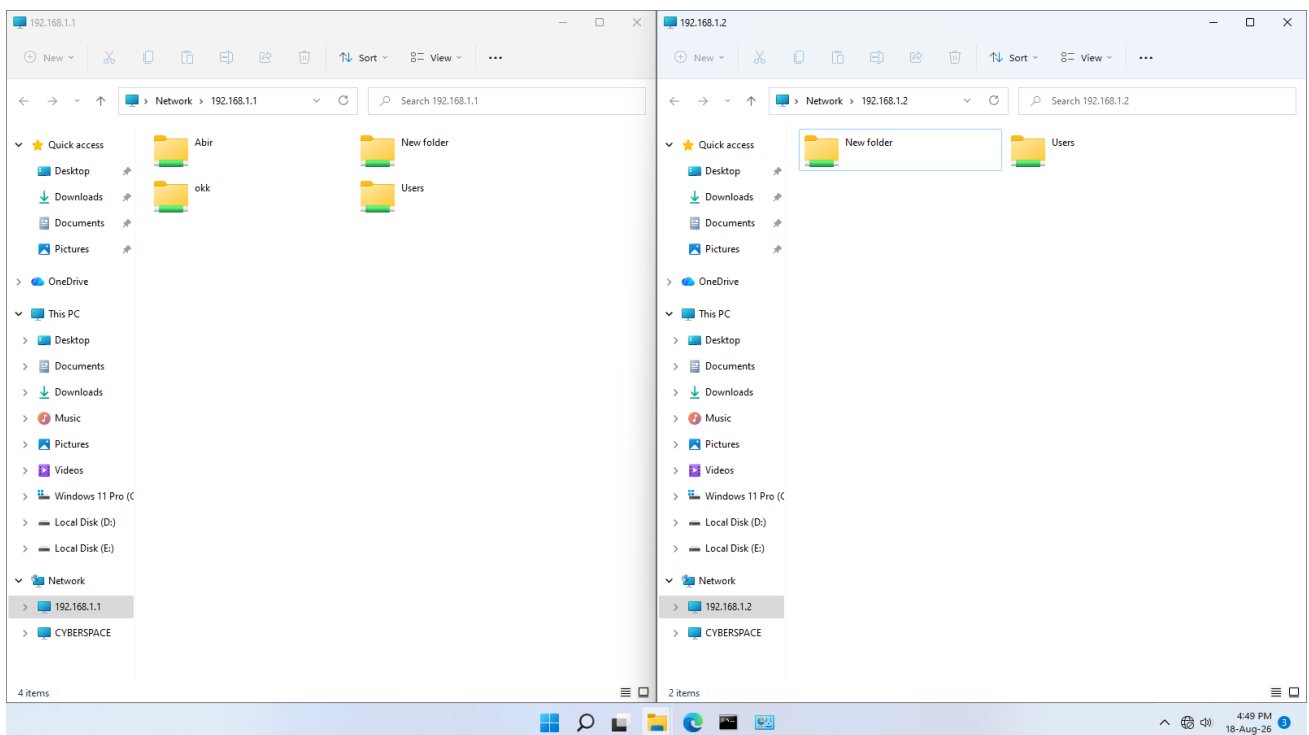
1. Open File Explorer on Computer 2
2. In the address bar, type: \192.168.1.1
3. Open the shared folder from Computer 1
4. Copy a test file to verify write access
5. Repeat this for Computer 3's shared folder: \192.168.1.3

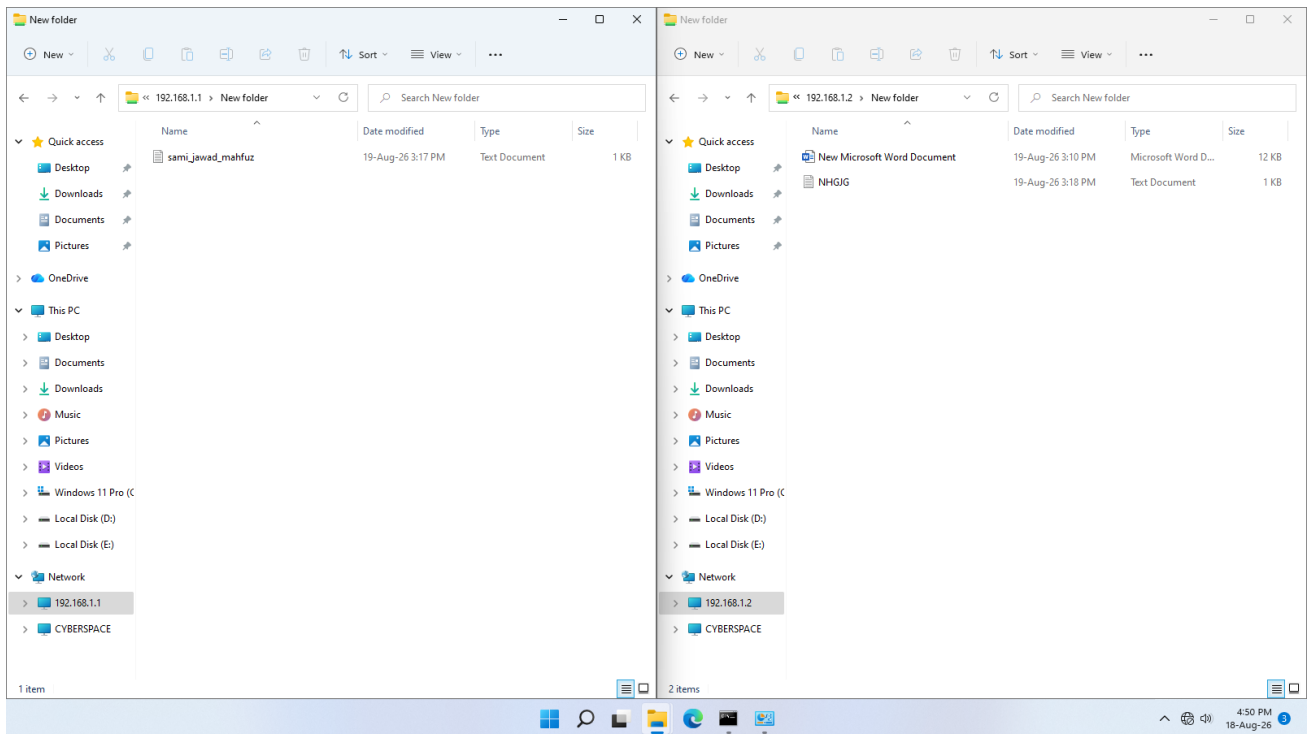




Step 11: Test File Sharing from Computer 3

1. Open File Explorer on Computer 3
2. Access Computer 1's shared folder: \\192.168.1.1
3. Access Computer 2's shared folder: \\192.168.1.2
4. Verify bidirectional file transfer capabilities
5. Document successful file transfers





Conclusion:

This experiment successfully demonstrated the establishment of a star topology network using a network switch and straight cables. Three computers were connected to a central switch, forming a star network configuration where all traffic flows through the switch. The initial configuration with private IP addresses (192.168.1.1, 192.168.1.2, 192.168.1.3) proved effective for local network communication. Ping tests confirmed successful connectivity between all computers, validating the physical connections and network configuration. File sharing was successfully enabled and tested among all three computers, demonstrating that the star topology effectively supports shared resources and collaborative file access. When the IP configuration was changed to simulate internet access (10.25.224.4, 10.25.224.5, 10.25.224.6), all computers successfully maintained connectivity both with each other and to the gateway, confirming that the star topology can seamlessly support both local and internet-based communication. The advantages of star topology were clearly demonstrated: centralized management through the switch, easier network expansion by adding computers to unused switch ports, improved network performance through dedicated switch pathways, and simplified troubleshooting due to the central connection point. This experiment provided practical understanding of how modern networks are structured and managed in enterprise and institutional environments.